

Electromagnetic Fields

Detailed Solutions • 2021–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

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PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electromagnetic Fields (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: IN	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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SOLUTIONS

Electrostatics

Topics

1. Coulomb's Law and Superposition Principle
2. Electric Field, Electric Field Intensity calculation in various Configurations: Due to Point, Line, Plane and Spherical Charge Distributions
3. Equipotential Surfaces and Electric Dipoles
4. Electric Flux Density and Gauss's Law
5. Divergence and Electric Potential
6. Electric Field Effect of Dielectric Medium- a)Boundary conditions b)method of image charges
7. Capacitance of Simple Configurations

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Unit Snapshot*

Stream: IN	**Questions:** 8	**MCQ:** 6	**MSQ:** 0	**NAT:** 2
1M: 4	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2021)–3M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M Ans: MTA ● ● ○

The electric field vector points in the direction of the maximum decrease of the electric potential function. The electric field intensity is related to the potential by $\bar{E} = -\nabla V$:

$$\bar{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}\right)$$

$$\bar{E} = -\left(\frac{x}{\sqrt{x^2+y^2+z^2}}\hat{x} + \frac{y}{\sqrt{x^2+y^2+z^2}}\hat{y} + \frac{z}{\sqrt{x^2+y^2+z^2}}\hat{z}\right)$$

At $(\hat{i}, \hat{j}, \hat{k})$ i.e., At $(1, 1, 1)$:

$$\bar{E} = -\frac{1}{\sqrt{3}}\hat{x} - \frac{1}{\sqrt{3}}\hat{y} - \frac{1}{\sqrt{3}}\hat{z}$$

$$\bar{F} = q\bar{E} = -\frac{1}{\sqrt{3}}\hat{x} - \frac{1}{\sqrt{3}}\hat{y} - \frac{1}{\sqrt{3}}\hat{z}$$

Comparing with $\bar{F} = a\hat{i} + b\hat{j} + c\hat{k}$:

$$a = -\frac{1}{\sqrt{3}}, \quad b = -\frac{1}{\sqrt{3}}, \quad c = -\frac{1}{\sqrt{3}}$$

MTA

Q1.26.2 MCQ 2026 2M Ans: A ● ○ ○

From Maxwell's first equation(Gauss's Law in differential form dictates that the divergence of the electric flux density field equals the volume charge density.):

$$\nabla \cdot \bar{D} = \rho_v \implies \rho_v = \varepsilon_o \nabla \cdot \bar{E}$$

Given the electric field vector components:

$$\nabla \cdot \bar{E} = \frac{\partial}{\partial x}(2x + 5y + 6z) + \frac{\partial}{\partial y}(5x + 4y + 10z) + \frac{\partial}{\partial z}(6x + 10y + 2z)$$

$$\nabla \cdot \bar{E} = 2 + 4 + 2 = 8$$

$$\rho_v = \varepsilon_o(8) = 8\varepsilon_o$$

$A) 8\varepsilon_o$

Q1.25.1 MCQ 2025 1M Ans: A ● ○ ○

The electric flux density $\bar{D}$ due to an infinite sheet of charge is independent of the distance from the sheet and depends solely on the surface charge density.
Given surface charge density $\rho_s = 10\,\text{C/m}^2$.

$$\bar{E} = \frac{\rho_s}{2\varepsilon}\hat{r}$$

$$\bar{E} = \frac{\rho_s}{2\varepsilon}\hat{c}$$

$$\varepsilon\bar{E} = \frac{\rho_s}{2}\hat{c}$$

$$\bar{D} = \frac{10}{2}\hat{c}$$

$$\bar{D} = 5\hat{c}$$

$A) 5\hat{c}$

Q1.25.2 NAT 2025 2M Ans: 4.00 to 4.40 ●●○

Electrostatic force between the plates is given by:

$$F_e = \frac{Q^2}{2\varepsilon A}$$

At Equilibrium, the mechanical spring restoring force balances the electrostatic attraction force.The restoring spring force is $F_s = kx$. At equilibrium position:

$$F_s = F_e \implies kx = \frac{Q^2}{2\varepsilon A}$$

$$Q^2 = 2\varepsilon Akx \implies Q = \sqrt{2\varepsilon Akx}$$

Substituting the given parameters:

$$Q = \sqrt{2 \times 8.854 \times 10^{-12} \times (100 \times 10^{-6})^2 \times 0.01 \times 10^{-6}}$$

$$Q = \sqrt{2 \times 8.854 \times 10^{-30}}$$

$$Q = \sqrt{1.7708 \times 10^{-29}} = 4.208 \times 10^{-14}\,\text{C}$$

$$\boxed{4.20}$$

Q1.24.1 MCQ 2024 1M Ans: A ●○○

The capacitance of a spherical shell arrangement depends on the geometric radii of the concentric spheres and the permittivity of the dielectric medium. The capacitance of a spherical capacitor with inner radius a and outer radius b is given by:

$$C = \frac{4\pi\varepsilon}{\left[\frac{1}{a} - \frac{1}{b}\right]} = \frac{4\pi\varepsilon ab}{(b-a)}$$

Given $a = x$ and $b = y$:

$$C = \left[\frac{4\pi\varepsilon xy}{y-x}\right]$$

$$\boxed{A)\ \frac{4\pi\varepsilon xy}{y-x}}$$

Q1.24.2 MCQ 2024 2M Ans: B ●●○

Given total current $I = \iint \bar{J} \cdot d\bar{A} = \iint ks^2\, \rho d\rho d\phi$ where $(s = \rho)$ represents the radial distance:

$$I = \iint k \cdot \rho^2 \cdot \rho d\rho d\phi = k \int \rho^3 d\rho \int d\phi$$

$$= k \left[\frac{\rho^4}{4}\right]_0^a [\phi]_0^{2\pi}$$

$$= \frac{k}{4}\left[a^4 - 0\right][2\pi]$$

$$I = \frac{\pi k a^4}{2}$$

$$\boxed{B)\ \frac{\pi k a^4}{2}}$$

Q1.22.1 NAT 2022 2M Ans: 33.3 to 33.4 ● ● ○

Given: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, $a = 10\,\text{cm} = 0.10\,\text{m}$ and $b = 15\,\text{cm} = 0.15\,\text{m}$.

The capacitance of concentric spherical conductors is

$$C = \frac{4\pi\varepsilon_0}{1/a - 1/b} = \frac{4\pi\varepsilon_0 ab}{b - a}.$$

Using the stated permittivity,

$$C = \frac{4\pi(8.854 \times 10^{-12})(0.10)(0.15)}{0.15 - 0.10} = 3.337950159 \times 10^{-11}\,\text{F} = 33.37950159\,\text{pF}.$$

Rounded to two decimal places, the answer is $\boxed{33.38\,\text{pF}}$.

Q1.21.1 MCQ 2021 1M Ans: B ● ○ ○

If an infinite line lies along the Z-axis, carrying a uniform charge density ρ_l C/m, then the electric field intensity in the cylindrical coordinate system (r, ϕ, z) is given by:

$$\bar{E} = \frac{\rho_l}{2\pi\varepsilon r}\hat{a}_r \quad \text{or} \quad \frac{\rho_l}{2\pi\varepsilon r}\hat{r}\ \text{V/m}$$

Therefore, the electric field intensity will be along the $\hat{r}$ direction only. The direction of the field is always radially outwards from the line charge.

$\boxed{\text{B) } \hat{r}}$

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SOLUTIONS

11

Magnetostatics

Topics

1. Biot-Savart's Law
2. Electromagnetic Induction and Faraday's Law
3. Magnetic Field of a current carrying loop
4. Ampere's Law and Curl
5. Lorentz Force
6. Inductance and Magnetomotive Force
7. Self and Mutual Inductance of Simple Configurations
8. Reluctance and Magnetic Circuits
9. Magnetic Flux and Boundary Conditions for magnetic field

Unit Snapshot*

Stream: IN	**Questions:** 6	**MCQ:** 2	**MSQ:** 0	**NAT:** 4
1M: 2	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2025)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 2M Ans: C ● ● ○

The differential force equation yields:

$$\bar{F} = Id\bar{\ell}\left(\frac{1}{\sqrt{6}}\hat{x} + \frac{2}{\sqrt{6}}\hat{y} + \frac{1}{\sqrt{6}}\hat{z}\right) \times (2\hat{x} + 2\hat{z})$$

Evaluating the cross product using determinants:

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 2 & 0 & 2 \end{vmatrix}$$

$$= \hat{x}\left(\frac{4}{\sqrt{6}} - 0\right) - \hat{y}\left(\frac{2}{\sqrt{6}} - \frac{2}{\sqrt{6}}\right) + \hat{z}\left(0 - \frac{4}{\sqrt{6}}\right)$$

$$= \frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}$$

$$\bar{F} = d\bar{\ell}\left(\frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}\right)$$

$$\frac{\bar{F}}{d\bar{\ell}} = \left(\frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}\right)$$

C) $\frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}$

Key Points

- The magnetic force on a differential current element is computed using the vector cross product of the direction vector and the magnetic field vector.

Q2.25.1 MCQ 2025 1M Ans: A ● ● ○

The magnetic force on a current-carrying conductor is maximized when the conductor is perpendicular to the magnetic field lines.

Given:

$$I = \frac{60}{0.5} = 120\,\text{A}$$

$$F = BIl\sin\theta \quad (\theta = 90^\circ)$$

$$F = BIl = 0.5 \times 1 \times 120 = 60\,\text{N}$$

Using the Right-Hand Rule, the direction is in the positive x-direction.

A) $60N$ in the direction of $\hat{x}$

Q2.23.1 NAT 2023 1M Ans: 5 ● ○ ○

Magnetic force per unit length between two parallel infinitely long current-carrying conductors, spaced 'd' apart is given by:

$$f = \frac{\mu I_1 I_2}{2\pi d}$$

$$f \propto \frac{1}{d}$$

$$\frac{f_2}{f_1} = \frac{d_1}{d_2} \implies f_2 = \left(\frac{d_1}{d_2}\right) f_1$$

Given: $d_1 = 2\,\text{cm}$ and $f_1 = 10\,\mu\text{N/m}$

$$d_2 = 2(d_1) = 2 \times 2\,\text{cm}$$

$$f_2 = \left(\frac{2}{2 \times 2}\right)(10)$$

$$\therefore\ f_2 = 5\,\mu\text{N/m}$$

$$\boxed{5}$$

Q2.22.1 NAT 2022 2M Ans: 200 ● ● ●

Given: Radius $= 0.1\,\text{m}, \omega = 20\,\text{rad/sec}, B = 2\,\text{T}$

$$T = \frac{2\pi}{\omega}, \quad V = -\frac{d\psi}{dt}$$

$$\frac{T}{2} = \frac{\pi}{\omega}$$

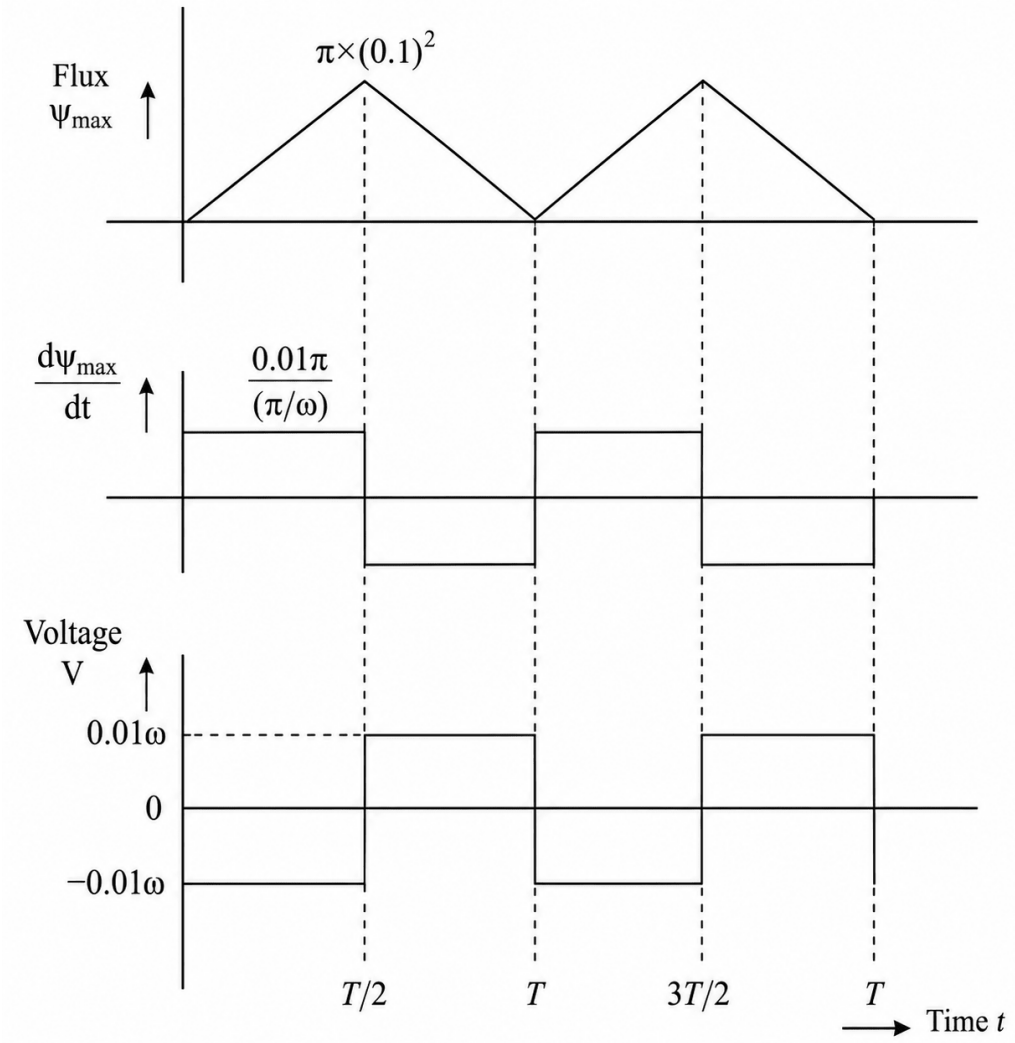


Fig. Solution Q2.22.1

$$V_{p-p} = 0.02\omega = 0.02 \times 20 = 0.4$$

$$I_{p-p} = \frac{0.4}{2} = 0.2 = 200 \times 10^{-3}$$

$$I_{p-p} = 200\,\text{mA}$$

$$\boxed{200}$$

Key Points

- Induced voltage is proportional to the time rate of change of magnetic flux linkage.
- Peak-to-peak values can be evaluated directly from the extrema transitions of the derivative waveforms.

Q2.21.1 NAT 2021 2M Ans: 2.7 to 2.9 ●●●

The net force (F) on the bar, which is subject to friction, is equal to the sum of the applied force (Electromagnetic Lorentz Force or F_{appl}) and the frictional force (F_{fr}).

$$F = F_{\text{appl}} - F_{\text{fr}}$$

(The negative sign is due to the frictional force opposing motion)

$$ma = IlB - kmg$$

Where: m : mass of bar 1 kg, I : current, $I = 10\,\text{A}$, l : length of bar $l = 0.5\,\text{m}$, $B = 1\,\text{T}$, k : Coefficient of friction $k = 0.1$, $g = 10\,\text{m/sec}^2$,

$$a = \frac{IlB}{m} - kg = \frac{10 \times 0.5 \times 1}{1} - 0.1 \times 10$$

$$a = 4\,\text{m/sec}^2$$

Acceleration, $a = 4\,\text{m/sec}^2$ From basics of Kinematics:

$$v^2 - u^2 = 2as$$

Where 's' is the distance travelled by the bar, $s = 1\,\text{m}$. As the bar is initially resting on the surface, its initial velocity $u = 0$.

$$v^2 = 2 \times (4) \times (1)$$

$$\therefore \text{ Speed, } v = 2.82\,\text{m/sec}$$

$$\boxed{2.82}$$

Q2.21.2 NAT 2021 2M Ans: 10 ●●○

Given: Toroid is made of CRGO with $\mu_r = 3000$, $\mu_0 = 4\pi \times 10^{-7}\,\text{H/m}$, $I = 5\,\text{mA}$, $N = 200$, Inner diameter, $D_1 = 10\,\text{cm}$ Outer diameter, $D_2 = 14\,\text{cm}$ Mean diameter:

$$D = \left(\frac{10 + 14}{2}\right) = 12\,\text{cm}$$

Mean radius, $R = 6\,\text{cm}$ Magnetic flux density in the core is given by:

$$B = \frac{\mu NI}{l}$$

Where l is the mean length of the toroid, $l = 2\pi R$.

$$B = \frac{\mu_0 \mu_r N I}{2\pi R}$$

$$B = \frac{4\pi \times 10^{-7} \times 3000 \times 200 \times 5 \times 10^{-3}}{2\pi \times 6 \times 10^{-2}}$$

$$B = \frac{2 \times 10^{-7} \times 3 \times 10^{3} \times 2 \times 10^{2} \times 5 \times 10^{-3}}{6 \times 10^{-2}}$$

$$B = 10\,\text{mT}$$

$$\boxed{10}$$

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Also in this Series

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Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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